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Spread of misinformation on social media: What contributes to it and how to combat it

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ABSTRACT

With the prevalence and increasing impact of misinformation on social media, numerous studies have explored the characteristics of misinformation. However, we lack a holistic understanding of what factors contribute to the viral spread of misinformation on social media and what strategies have been proposed to combat the phenomenon. Given the importance of these questions to misinformation research, this study aims to (a) provide a systematic and structured overview of the factors that influence the spread of misinformation by analyzing the four vital elements of information communication, namely, source, message, context, and receiver and (b) summarize the current state of research on strategies against the spread of misinformation on social media from various perspectives and discuss their advantages, disadvantages, and effectiveness. Following the process of a systematic literature review, this study identifies and analyzes 423 relevant articles. Finally, we highlight research gaps in the existing literature and recommend directions for future research.

1. Introduction

Although the spread of misinformation is as old as human history, social media has changed the game by enabling people to generate misinformation easily and spread it rapidly in an anonymous and decentralized fashion (Del Vicario et al., 2016; Wu et al., 2016). The impact of misinformation can be destructive to various aspects of our lives, from public health and politics to climate change and economic issues. For instance, the viral spread of misinformation during the COVID-19 pandemic has raised various psychological issues among the public and led them to adopt inappropriate protective measures (Dong et al., 2020; Pian, Chi, & Ma, 2021). In politics, beyond influencing citizens' electoral behaviors, the potential influences of the spread of misinformation include increasing political fragmentation and lowering general trust in the government (Garrett, 2019; Lazer et al., 2018). Moreover, pseudoscience about climate change has confused the public and hindered citizens' responses to mitigation policies (Cook, Ellerton, & Kinkead, 2018; Treen et al., 2020). Economy-related topics are also a popular target of misinformation with notable effects, such as stigmatizing the reputation of corporations and affecting consumers' purchase intentions (Berthon & Pitt, 2018; Di Domenico, Sit, Ishizaka, & Nunan, 2021). The prevalence and negative impact of misinformation on social media call for a careful and systematic review of the relevant literature focusing on understanding factors that contribute to the spread of misinformation and developing strategies to confront this spread.

1.1. Misinformation terminology

In academic literature, various terms or concepts are related to misinformation; they mainly differ in terms of intention (the degree to which the author deliberately misleads the audience) and facticity (the degree to which the content relies on facts). Most of them can be classified into the following three major types of information disorder (see Fig. 1, Wardle & Derakhshan, 2017): (1) "Misinformation", which refers to false information that is spread without the intent to mislead, such as unwittingly disseminating a story that contains outdated information; (2) "Disinformation", which refers to false information that is deliberately produced to cause harm, such as reporting a news containing manipulated statistics; and (3) "Malinformation", which refers to true information propagated to cause harm, such as circulating a statement outside of its original context to mislead the audience. Other related concepts are also frequently used in the literature. For instance, rumor

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refers to circulating stories or statements, the facticity of which is unverified and may turn out to be true or partly or entirely false (Zubiaga et al., 2016). Fake news refers to fabricated information that intends to deceive by mimicking the form of legitimate news (Lazer et al., 2018).

Because the intent behind information is hard to discern, to simplify this complicated phenomenon, as many previous researchers did, we do not draw a sharp distinction between misinformation and other related concepts in this review, and we use the term misinformation to collectively represent any information that turns out to be false.

1.2. Research gap on the review of misinformation spread

The complex and hyper-dynamic information environment introduced a new era in studying misinformation over the recent decade that considered influences from social media and social networks. At the same time, numerous reviews emerged to synthesize research regarding the spread of misinformation from different perspectives. A summary of recently published reviews is shown in Table 1. The major inadequacies of these reviews can be summarized as follows.

First, an overarching synthesis of the diverse factors influencing the spread of misinformation is lacking. Baptista and Gradim (2020) summarize factors behind the viral spread of misinformation from two perspectives: the motivation of consumers and the content presentation of online misinformation. Some research focuses on psychological drivers (Ecker et al., 2022; Pennycook & Rand, 2021), including cognitive, social, and affective factors. However, factors associated with the social media context and social network structure that also significantly contribute to the spread of misinformation were not systematically reviewed.

Second, many studies have reviewed the methods of detecting misinformation (Sharma et al., 2019; Zhang & Ghorbani, 2020), but less attention has been paid to synthesizing misinformation-countering strategies, such as what to do with the posts containing misinformation that escape the detection of fact-checking systems and how to effectively articulate content that corrects misinformation on social media to reduce its impact. Pian et al. (2021) summarized strategies proposed against misinformation, but most of these strategies are based on experts' commonsense or implications inferred from misinformation studies, whereas few are based on empirical studies on misinformation-countering strategies.

Third, previous research has tended to focus on specific topics of misinformation, such as politics (Bryanov & Vziatysheva, 2021; Pennycook & Rand, 2021; Tucker et al., 2018), health (Wang et al., 2019; Melchior & Oliveira, 2022), COVID-19 (Janmohamed et al., 2021; Pian et al., 2021), business (Di Domenico, Sit, et al., 2021), and climate change (Treen et al., 2020). Limited work has focused on misinformation with broader topics. Therefore, previous review research cannot comprehensively depict the factors affecting the virality of misinformation and countermeasure strategies against this phenomenon on social media.

Fourth, the scope of articles reviewed by most systematic reviews is limited to those published in the last five years, and articles published before 2016 were overlooked. Additionally, some systematic reviews focused on specific types of online misinformation, especially fake news (Baptista & Gradim, 2020; Celliers & Hattingh, 2020; Di Domenico, Sit, et al., 2021; Pennycook & Rand, 2021; Wu, Ngai, Wu, & Wu, 2022), and used limited search queries to acquire relevant articles, which omitted other types of misinformation that can be influential on social media.

1.3. Motivations and contributions of this study

The motivations of this study can be summarized as follows. (1) The extant reviews of misinformation studies are insufficient to establish a comprehensive understanding of factors affecting the spread of misinformation. Some important relevant perspectives, such as information context and social network structure, are also demonstrated in this study to generate an overall comprehension of the spread of misinformation on social media. (2) Despite methods of detecting misinformation, studies on misinformation-countering strategies are diverse in terms of their targets, such as the source and message content. It is necessary to summarize the different types of strategies proposed and their effectiveness, as supported by empirical studies. By presenting a comprehensive view of strategies against the spread of misinformation on social media, this research review can provide insights for academic research and practical application. (3) Different from previous literature surveys that target specific topics or types of misinformation, in this work, we review misinformation studies not limited to those focusing on a particular topic or type of misinformation. Additionally, we trace relevant articles back to 2010. Therefore, we can provide readers with a more comprehensive view of the factors influencing the virality of misinformation and existing solutions to confront it.

The major contributions of this study can be summarized as follows. (1) We present a detailed discussion of factors that influence the spread of misinformation on social media from various perspectives. (2) We present the categorization of proposed strategies for fighting misinformation and discuss the effectiveness, advantages, and disadvantages of these strategies. (3) Based on this review, we identify several key challenges and recommend potential directions for future research.

2. Methods

2.1. Database search strategy

In this section, we list the databases and queries used to search for relevant articles and present the selection criteria based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (PRISMA) (Moher, Liberati, Tetzlaff, Altman, & Group, 2009), wherein we illustrate a step-by-step systematic approach for selecting the articles.

We searched the Web of Science (WoS) Core Collection and Scopus for journal articles and conference papers written in English and published between January 2010 and September 2021. The search terms are shown in Table 2. This yielded 2148 records from WoS and 2853 records

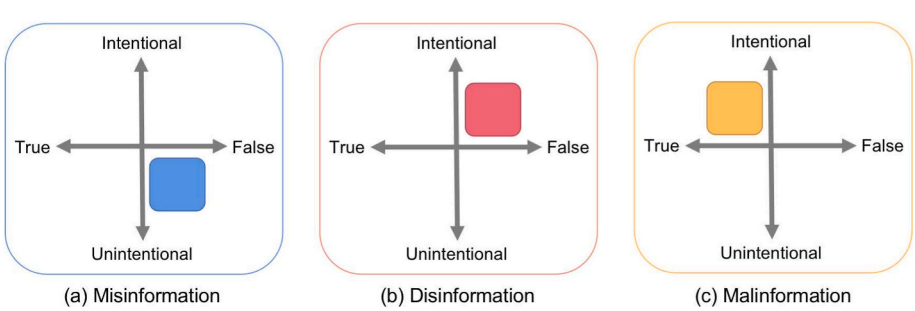

Fig. 1. Difference between misinformation, disinformation, and malinformation.

Table 1
Literature review studies on misinformation spread.

Authors	Focus	Target topic of misinformation	Type of misinformation	Scope of reviewed articles
Tucker et al. (2018)	Factors (social media usage and political polarization)	Politics	Any type of information that turns out to be false.	not report
Sharma et al. (2019)	Detection methods	General misinformation	Any type of information that turns out to be false.	not report
Zhang & Ghorbani (2020)	Characteristics and detection methods	General misinformation	Any type of information that turns out to be false.	not report
Celliers and Hattingh (2020)	Factors (social factors, cognitive factors, political factors, financial factors, and malicious factors)	General misinformation	Fake news	Including 38 related articles published between 2016 and 2019
Baptista and Gradim (2020)	Factors (individual motivation and message presentation)	General misinformation	Fake news	Including 52 related articles published between 2016 and 2020
Treen et al. (2020)	Factors (human and platform factors); Strategies (education, inoculation, technological solutions, response, and regulation)	Climate change	Climate change misinformation/skepticism/ contrarianism/denial	not report
Bryanov and Vziatysheva (2021)	Factors (message presentation, individual factors, and accuracy-promoting interventions)	Politics	Any type of information that turns out to be false.	Including 26 related articles published between 2016 and 2020
Pian et al. (2021)	Factors (social media usage, health literacy, fast publication process); Strategies (countermeasures for a low level of health literacy, social media platforms, governments, risk communication, and health information needs)	COVID-19	Any type of information that turns out to be false.	Including 251 related articles published up to January 2021
Di Domenico, Sit, et al. (2021)	Factors (attitudes, dissemination process, and spreading channel)	Business	Fake news	Including 117 related articles published up to November 2019
Pennycook and Rand (2021)	Factors (political motivations, reasoning, and heuristics)	General misinformation	Fake news	not report
Wu, Ngai, Wu, & Wu (2022)	Factors (individual factors, FNI information, debunking information, the internet media, and information environment)	General misinformation	Fake news	Including 202 related articles published up to October 2021
Janmohamed et al. (2021)	Strategies (misinformation correction)	COVID-19	Any type of information that turns out to be false.	Including 12 related articles published between 2020 and 2021
Ecker et al. (2022)	Factors (cognitive, social, and affective factors); Strategies (misinformation correction)	General misinformation	Any type of information that turns out to be false.	not report

Table 2
Databases and searching queries.

Database name	Query string
Web of Science Core Collection	TOPIC: ((misinformation OR disinformation OR "fake news" OR "false information" OR rumo* OR "misleading information") AND ("social media" OR "social network") AND (spread* OR shar* OR diffus* OR disseminat* OR propagat* OR broadcast* OR circulat*))
Scopus	TITLE-ABS-KEY (misinformation OR disinformation OR "fake news" OR "false information" OR rumo* OR "misleading information") AND ("social media" OR "social network") AND (spread* OR shar* OR diffus* OR disseminat* OR propagat* OR broadcast* OR circulat*)

from Scopus. After removing duplicates, 3214 records were left.

2.2. Selection process and eligibility criteria

Screening the titles and abstracts of 3214 records, articles not involving social media, unrelated to the spread of misinformation, and not written in English were excluded. At this and the subsequent stages, we also excluded articles focusing on computational approaches to detect misinformation automatically, which have already been exhaustively reviewed by many researchers (e.g., Shahid et al., 2022; Sharma et al., 2019; Zhang & Ghorbani, 2020).

Next, we classified the remaining 1152 records into four categories, namely, factor-related, strategy-related, both factor- and strategy-related, and not related, based on an analysis of abstracts. To assess the intercoder reliability, we randomly selected 450 samples for training. After three rounds of annotations and discussions, we reached acceptable intercoder reliability with a Fleiss kappa value of 0.805 (Cohen, 1960). After annotation training, each author coded 351 records and discussed the uncertain ones. This left 459 potentially eligible articles, which were subject to full-text analysis, after applying the following eligibility criteria:

- Including empirical studies that investigate factors affecting the spread of misinformation on social media
- Including theoretical studies that propose mathematical models to simulate the spread of misinformation and consider specific factors affecting the spread of misinformation on social media
- Including empirical studies that propose a specific approach or examine the effectiveness of a specific approach to combat the spread of misinformation on social media
- Including theoretical studies of mathematical models that propose an approach to minimize the spread of misinformation on social media and verify the effectiveness of the approach
- Excluding studies on the spread of misinformation not in the context of social media
- Excluding articles that only focus on investigating the characteristics or impacts of the spread og misinformation on social media but do not verify what factors influence this spread
- Excluding articles that mention the countermeasures against misinformation only when discussing the implication of their study

Finally, 423 articles were selected for data analysis and synthesizing. The detailed screening and selection process is shown in Fig. 2.

2.3. Data synthesizing

Fig. 3 shows the number of articles eligible for the literature review classified according to the focus of the study and year of publication.

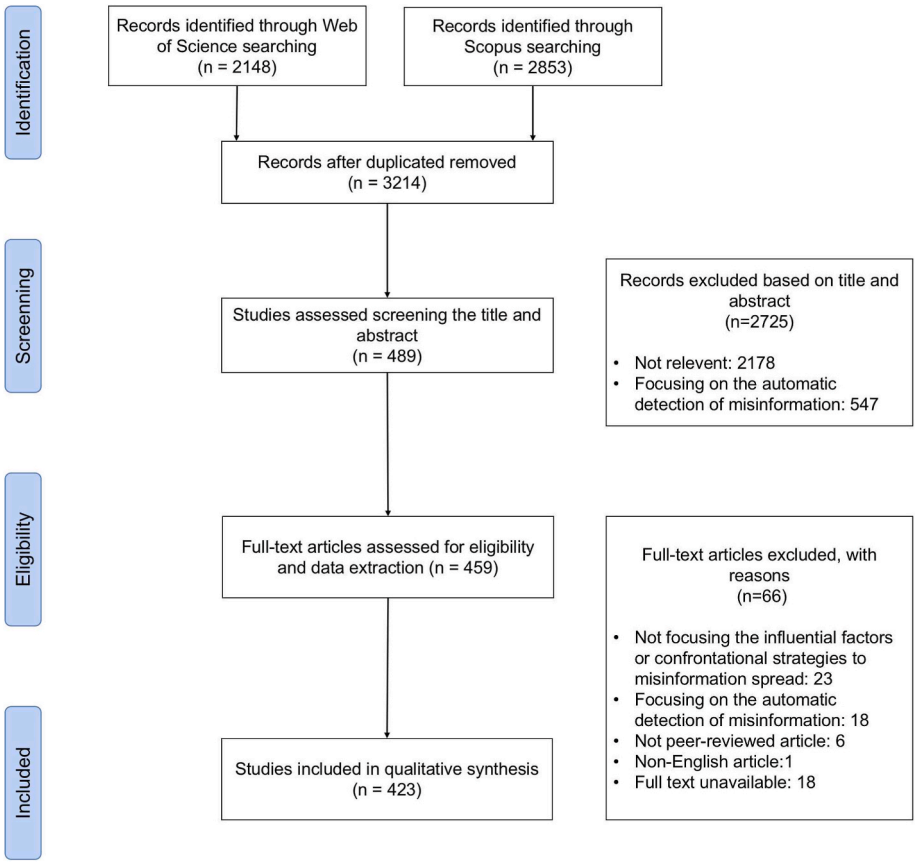

Fig. 2. Article screening and selection.

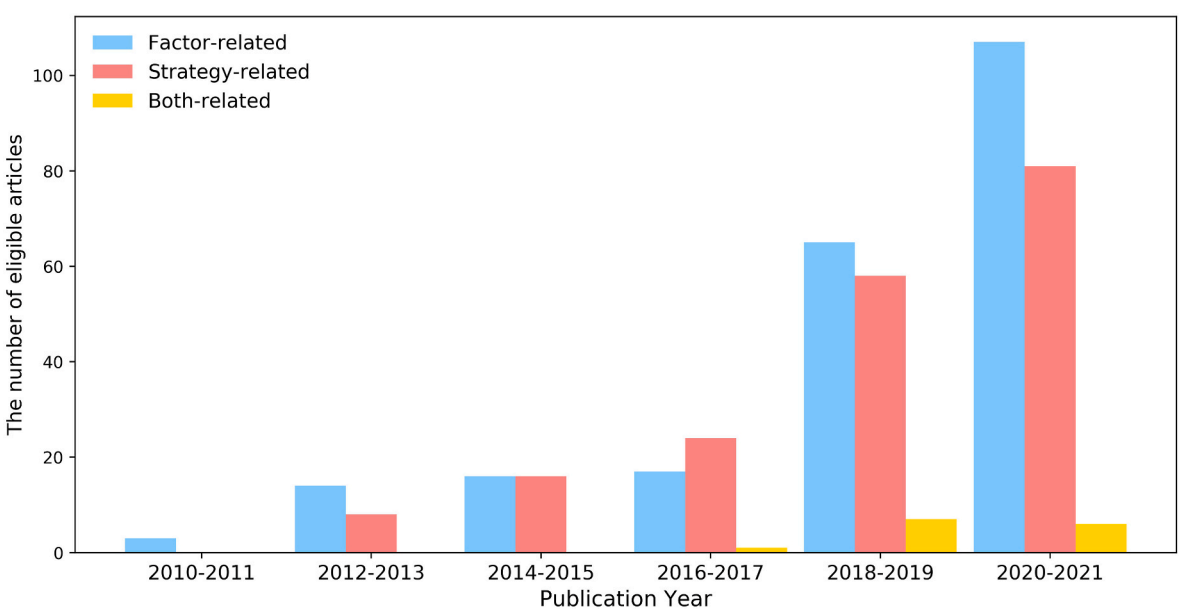

Fig. 3. Final number of articles eligible for the review.

From 2010 to 2011, only three articles regarding the factors affecting the spread of misinformation were eligible for review. The publication of articles addressing strategies against the spread of misinformation on social media began in 2012. During the last four years (2018–2021), the publication of relevant articles has increased significantly.

Then, a content analysis approach was applied to synthesize the findings from different articles. A coding framework was developed using an iterative coding process. For example, if one particle factor was identified from an article during the coding, we tried to assign it to a specific category within the factor framework. A new subcategory was created if the current categories were unsuitable until all the factors were categorized. The same procedure was performed for identifying strategies to confront misinformation from the selected articles. The details of the coding result are provided in the Supplementary material.

3. Results

Berlo's SMCR model of communication (Berlo, 1960; Schulz & Cobley, 2013, p. 402) highlights four fundamental components of a communication process, namely, the source/sender, message, context/channel, and receiver (as shown in Fig. 4). In the following subsections, we summarize the identified factors of misinformation spread on social media and the strategies researchers have explored to fight against this spread in these four categories. Specifically, in the process of misinformation spread on social media, "source" refers to the originator or sender of the message containing misinformation. "Message" is the package of misinformation and can be sent in various forms (e.g., text, image, video, etc.). "Context" denotes the information environment wherein misinformation is spread (e.g., the social media channel, the receiver's social network, etc.). "Receiver" refers to the person who receives and subsequently processes the misinformation-containing message.

3.1. Key factors driving the spread of misinformation on social media

Aligning with the four communication elements according to the SMCR model, we summarize the factors that may affect the spread of misinformation on social media in four categories, i.e., source-, message-, context-, and receiver-related factors. The joint forces of influential source, susceptible user, well-crafted message, and conductive environment lead to the viral spread of misinformation (as shown in Fig. 5).

3.1.1. Source-related factors

Defining and evaluating the source becomes particularly difficult when information travels through social media because any user can be a content publisher (Wang et al., 2019). From the audience's perspective, the source could be defined as either the one from whom they receive the message or the originator of the misinformation content. From previous research, a range of source-related characteristics showed the influence of the source on the misinformation spread on social media, including credibility, ambiguity, popularity, similarity, and self-interest of source.

Most previous studies highlight the importance of source credibility or senders' trustworthiness as factors affecting social media users' belief in misinformation content (Bae, 2020; Kim, Bock, Sabherwal, & Kim, 2019; Nadarevic, Reber, Helmecke, & Köse, 2020) and subsequent susceptibility to sharing misinformation posts (Nadarevic et al., 2020; Seah & Weimann, 2020; Liu, Burton-Jones, & Xu, 2013; Liu, Burton-Jones, & Xu, 2014; Buchanan & Benson, 2019; Buchanan, 2020). Another source-related factor is source ambiguity, which reflects the extent to which the spreader of misinformation understands the source of a message and its trustworthiness (Oh, Agrawal, & Rao, 2013; Liu, Burton-Jones, & Xu, 2014). If there is enough importance and little ambiguity, or slight importance but a greater degree of ambiguity, the misinformation will spread (Cheng, 2019). Source popularity, measured by the number of followers, can affect the propagation of misinformation content (Gupta, Lamba, & Kumaraguru, 2013; Mikhaeil & Mougy, 2020). Additionally, similarity and the self-interest of the source contribute to the spread of misinformation (Lee & Shin, 2021). That means individuals are more susceptible to misinformation from sources that show similar interests or viewpoints that are similar to theirs.

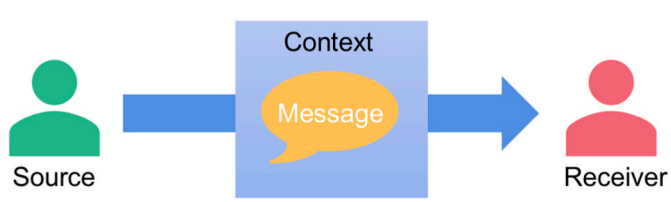

Fig. 4. SMCR model of communication.

Information from third-party sources, such as grassroots users' comments, is more likely to be accepted or shared than comments from the stakeholders themselves.

3.1.2. Message-related factors

It is widely acknowledged that message-related features would affect how the information is perceived and distributed. Through the review, the following major message-related factors are identified: presentation formats, language styles, and psychological cues.

Presentation formats. Various studies investigated the effect of messages' presentation format on the spread of misinformation. For example, images or graphic elements are found to attract the attention of readers and motivate them to share the misinformation on social media (Burel, Farrell, Mensio, Khare, & Alani, 2020; Hameleers, Powell, Van Der Meer, & Bos, 2020; Nanni et al., 2020). However, the influence of visual content might decrease and even vanish when other factors, such as source credibility, are considered. For instance, Nadarevic et al. (2020) investigated how the presentation format (with or without a picture), source credibility (expert, layperson, or no source), and repetition (repeated message exposure) jointly affect users' propensity to share misinformation on social media. They found no effect of presentation format but strong effects of source credibility and repetition (Nadarevic et al., 2020). Additionally, when misinformation is presented in a source-primacy format, recipients may be less likely to share the post (Di Domenico, Nunan, Sit, & Pitardi, 2021). Moreover, the presence of interactive elements, such as URLs, mentions, and hashtags, may contribute to the viral sharing of misinformation on social media (Kumar, Huang, Cox, & Carley, 2021; Tanaka et al., 2014).

Language styles. Storytelling may enhance fluency and contribute to the spread of misinformation (Chen, Xiao, & Mao, 2021; Dahlstrom, 2021; Lee & Shin, 2021). Studies consistently indicate that it is difficult for people to identify falsehoods within narratives, and they usually accept these falsehoods at face value (Dahlstrom, 2021). Because narratives or storytelling can trigger both rational and emotional processes in readers (Fisher, 1987), they are easier to comprehend, and once they are integrated into the existing knowledge structure of readers, they can be hard to correct (Green & Donahue, 2018; Lee & Shin, 2021). Furthermore, factual statements or evidence are also found to be exploited to increase the strength of arguments and credibility of misinformation content (Igwebuike & Chimuanya, 2021; Lee & Shin, 2021; Cheong & Babcock, 2021; Chen, Xiao, & Mao, 2021). The higher the perceived credibility of misinformation is, the stronger the social media users' willingness to forward the message is (Halpern, Valenzuela, Katz, & Miranda, 2019; Kim, Bock, et al., 2019; Seah & Weimann, 2020). Additionally, to increase the fluency of the message and deepen the audience's memory, content repetition may contribute to the perceived accuracy of misinformation and the likelihood of its going viral on social media (Chen, Xiao, & Mao, 2021; Lee & Shin, 2021; Pennycook, Cannon, & Rand, 2018).

Psychological cues. Instances of misinformation content differ in their psychological motivators and intended effects. For instance, emotional language and appeals to emotional reactions are often used to frame misinformation and are found to contribute to the viral spread of misinformation (Baptista & Gradim, 2020; Chen, Xiao, & Mao, 2021; Igwebuike & Chimuanya, 2021; Kumar et al., 2021; Nazar & Pieters, 2021; Song et al., 2019a; Vosoughi et al., 2018; Yeo and McKasy, 2021; Zhou et al., 2021). Brady, Gantman, and Van Bavel, (2020) explained the psychological processes behind the viral spread of moral and emotional content, that is, such content has a high potential to capture attention, which is found to be associated with forwarding behavior on social media. Additionally, some misinformation posts strike panic while others offer hope. Chua and Banerjee (2018) found that panic-inducing misinformation is more likely to be believed and shared than those appealing to hope. Moreover, various studies indicate that social media users have a greater propensity to be attracted to content that confirms their identity or prior beliefs regardless of the posts'

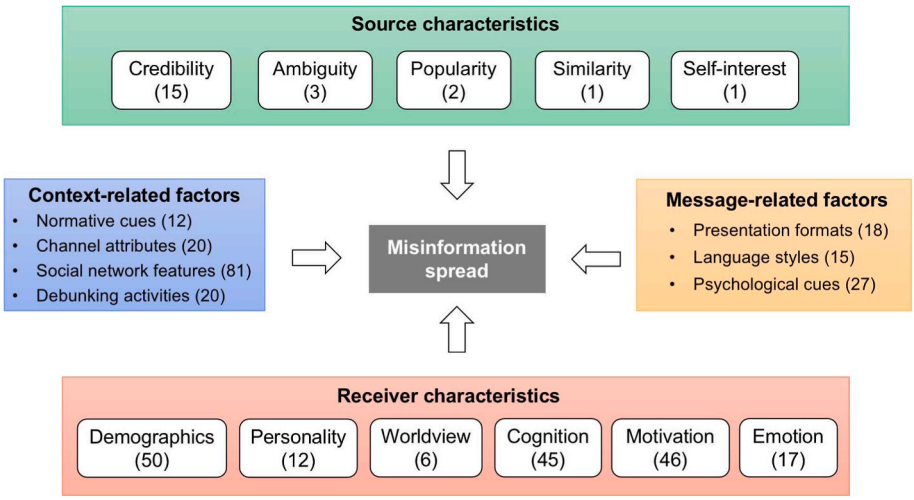

Fig. 5. Factors affecting the spread of misinformation (236 related articles in total, of which 108 are about the combination of different types of factors).

factual basis (Bowyer & Kahne, 2019; Buchanan, 2020; Kim, Bock, et al., 2019; Krafft & Donovan, 2020; Neyazi & Muhtadi, 2021). Whether a misinformation-containing message posted on social media indicates a feeling of involvement (the cue of personal involvement) could predict its likelihood to be shared (Chua & Banerjee, 2018; Koohikamali & Kim, 2015; Liu et al., 2014; Nazar & Pieters, 2021; Oh et al., 2013; Park & Rim, 2020; Song et al., 2019a). Furthermore, interesting, eye-catching, or controversial content can increase the spread of misinformation (Abdullah, Nishioka, Tanaka, & Murayama, 2015; Chen, Sin, Theng & Lee, 2015a, 2015b; Shen et al., 2021).

3.1.3. *Receiver-related factors*

Some individuals may be more vulnerable to misinformation than others. In this section, we reviewed individual-related factors from the perspective of the misinformation receiver. Generally, the following six types of factors are identified: demographics, personality traits, worldviews, motivations, cognition, and emotional state.

Demographic features. People with different demographics, such as education level, age, gender, geographic location, and media habits, would respond to misinformation differently. For example, less educated people were found to be more likely to believe misinformation (Lai, Xiong, Jiang, Sun, & He, 2020; Melki et al., 2021; Schaewitz et al., 2020; Scherer et al., 2021) and share misinformation on social media (Buchanan, 2020; Chen et al., 2015a; Schaewitz et al., 2020). A possible explanation is that individuals with a longer period of education are more likely to possess the knowledge and ability to detect misinformation (Hui, Zhou, Lü, & Li, 2020; Zhang et al., 2016). Similarly, a lower level of skepticism and higher level of acceptance of misinformation were observed among older people (Bapaye & Bapaye, 2021; Blanco-Herrero, Amores, & Sánchez-Holgado, 2021; Loos & Nijenhuis, 2020; Rampersad & Althiyabi, 2020) and they were more likely to share misinformation (Guess, Nagler, & Tucker, 2019; He, Xiong, & Lai, 2019; Osmundsen, Bor, Vahlstrup, Bechmann, & Petersen, 2021). Additionally, people under 40 years of age were more likely to share misinformation if accompanied by pictures (Tanțău et al., 2019). How gender differs in susceptibility to misinformation is inconsistent. For instance, some studies found that males were more likely to trust misinformation (Buchanan, 2020; Filkuková, Ayton, Rand, & Langguth, 2021; Laato, Islam, Islam, & Whelan, 2020), whereas some research showed that females had a greater tendency to trust or share misinformation (Melki et al., 2021; Raj & Goswami, 2020; Chen et al., 2015a; Li et al., 2017; Chen et al., 2015b). Furthermore, during crises, individuals' geographical proximity to the crisis was negatively associated with misinformation-sharing behavior (Chua, Pal & Goh, 2020; Huang, Starbird, Orand, Stanek, & Pedersen, 2015; Valecha et al., 2019), because people who are physically close to the crisis would perceive high personal involvement with the misinformation (Chua et al., 2020). Individuals' media habits are relevant to their misinformation-sharing behaviors on social media. For example, people with high levels of social media use (Blanco-Herrero et al., 2021; Buchanan, 2020; He, Yang, Xiong, & Lai, 2019; Su, 2021), smartphone use (Raj & Goswami, 2020), and online trust (Laato et al., 2020; Melki et al., 2021; Talwar et al., 2019) are more likely to believe and spread misinformation.

Personality traits. Recently, there has been an increasing interest in exploring the personality traits that lead to the spread of misinformation on social media (Buchanan & Benson, 2019; Calvillo, Garcia, Bertrand, & Mayers, 2021; Chen, 2016; Chen & Sin, 2013; Gumelar, Akbar, & Erik, 2018; Lai et al., 2020; Lawson & Kakkar, 2022; Mikhaeil & Mougy, 2020; Williams & Dupuis, 2020). For instance, Chen (2016) showed that neurotic people were less likely to share misinformation but open-minded people might have wider interests and were more likely to share misinformation. Lai et al. (2020) found a contrary conclusion that people high in neuroticism were likely to fall for misinformation. Moreover, they found an effect of extroversion on belief in misinformation and a moderating effect of gender. The work by Buchanan and Benson (2019) suggests that agreeable people were less likely to engage with a misinformation post because it was inconsistent with their general friendly disposition; however, they did not find any significant relationship between the other four types of personality traits (i.e., openness, conscientiousness, extroversion, and neuroticism) and misinformation engagement. Conversely, Calvillo et al. (2021) found that agreeable, conscientious, and open-minded people are susceptible to misinformation.

Worldviews. Research has also shown that individuals' worldviews, belief systems, and ideologies play an important role in people's responses to misinformation (Freelon et al., 2022; Grinberg, Joseph, Friedland, Swire-Thompson, & Lazer, 2019; Guess et al., 2019; Halpern et al., 2019; Islam, Laato, Talukder, & Sutinen, 2020; Jost, van der Linden, Panagopoulos, & Hardin, 2018; Lawson & Kakkar, 2022; Lutzke, Drummond, Slovic, & Árvai, 2019; Waruwu et al., 2021). For instance, conspiracy mentality (a belief system related to prejudice against powerful groups and perceiving them as covertly responsible for negative political or economic events) is found to be positively correlated with believing and sharing misinformation (Halpern et al., 2019), whereas one's religious belief has a negative effect on sharing misinformation (Islam et al., 2020). Several studies indicate that people with a conservative political ideology are more likely to accept and spread misinformation than people with a liberal political ideology (Grinberg et al., 2019; Guess et al., 2019; Halpern et al., 2019; Jost et al., 2018; Lutzke et al., 2019). Based on their exploration of the interaction effect

between political ideology and conscientiousness, Lawson and Kakkar (2022) suggest that the misinformation sharing is driven by low conscientiousness conservatives and there is no difference between liberals and conservatives when one is at a high level of conscientiousness.

Motivations. A range of psychological motivations drive people to fall for and spread misinformation, such as socializing (Apuke & Omar, 2020b, 2021; Chen & Sin, 2013; Chen, Sin, Theng & Lee, 2015a, 2015b; Shen et al., 2021), self-promotion (Apuke & Omar, 2020; Islam, Mäntymäki, & Benbasat, 2018; Islam et al., 2020; Shen et al., 2021), altruism (Apuke & Omar, 2020b, 2021; Seah & Weimann, 2020), and information seeking (Apuke & Omar, 2021; Chen & Sin, 2013; Duffy, Tandoc, & Ling, 2020; Shen et al., 2021). The socializing motivation reflects the motive to share information to develop and enhance relationships with others on social media (Apuke & Omar, 2020b; Chen et al., 2015a; Oh & Syn, 2015). Socializing is considered as an important psychological motivation for sharing misinformation because sharing can be an excellent way to connect and build relationships with others (Apuke & Omar, 2020b, 2021; Chen & Sin, 2013; Chen, Sin, Theng & Lee, 2015a, 2015b; Shen et al., 2021). Self-promotion or self-enhancement refers to the willingness to present oneself to other people as expert or knowledgeable (Apuke & Omar, 2020b; Islam et al., 2020; Tellis et al., 2019). Those who are motivated by self-promotion tend to seek others' following, liking, and sharing, thereby, prompting the spread of misinformation (Apuke & Omar, 2020b). Altruism refers to the act of sharing information to help others without expecting any benefit in return (Plume & Slade, 2018). Several studies found that altruism is positively correlated with sharing misinformation related to COVID-19 or food safety (Apuke & Omar, 2020b, 2021; Seah & Weimann, 2020). A possible explanation for this finding could be that during the pandemic or other emergent situations, people want to immediately share information they believe might help others, and in doing so, they barely consider the truthfulness of the information (Apuke & Omar, 2020b). The information-seeking motivation refers to the level to which individuals wish to be "in the know" (Apuke & Omar, 2021; Chen & Sin, 2013; Duffy et al., 2020). As a way of fact finding and information seeking, sharing misinformation on social media could resolve the sender's feeling of uncertainty and restore control of a situation (Shen et al., 2021).

Knowledge and cognition. Individuals' low level of issue knowledge or scientific knowledge (Bowyer & Kahne, 2019; Huo & Chen, 2020; Osmundsen et al., 2021; Park & Rim, 2020; Pennycook, McPhetres, et al., 2020; Vegetti & Mancosu, 2020; Wang et al., 2021) and cognitive ability (Ahmed, 2021; Bansal & Weinschenk, 2020; Brashier & Schacter, 2020; Lee et al., 2020; Pennycook, McPhetres, et al., 2020) would affect their susceptibility to misinformation on social media. In particular, individuals with a high level of critical thinking are more likely to process information through an elaborated route, rather than a heuristic route, and hence be less vulnerable to misinformation (Buchanan, 2020). Unfortunately, the current information environment wherein inaccurate and biased information is prevalent has, in turn, amplified cognitive biases and limitations (Barzilai & Chinn, 2020). Additionally, lazy thinking (Pennycook & Rand, 2019a), perceived information overload (Bermes, 2021; Laato et al., 2020), and strong emotions (Tiedens & Linton, 2001; Lee, Kim, & Kim, 2021; Wang et al., 2020; Yeo and McKasy, 2021) can impair people's cognitive capacity to process information and lead to an increased likelihood of sharing misinformation. People with strong epistemic beliefs were found to be more likely to share misinformation conveyed by text than by images (Chua & Banerjee, 2017). Another critical factor is information literacy, which consists of knowledge and skills related to identifying, seeking, analyzing, and utilizing information for the issue at hand (ACRL, 2000). People with a high level of information literacy might be more aware of the surrounding information and be better at evaluating the credibility of misinformation shared through social media (Di Domenico, Sit, et al., 2021). Furthermore, various types of information-related literacy (e.g., media literacy, digital literacy, and health literacy) have emerged from different domains and have been found to be correlated with users' susceptibility to misinformation (Brashier & Schacter, 2020; Buchanan, 2020; Melki et al., 2021; Oh & Lee, 2019; Osmundsen et al., 2021; Raj & Goswami, 2020; Song et al., 2019a, 2019b; Soto-Vásquez et al., 2021; Syam & Nurrahmi, 2020).

Emotional state. Existing studies (Aswani, Kar, & Ilavarasan, 2019; Dong et al., 2020; He et al., 2019; Lee et al., 2021; Liu et al., 2014; Oh et al., 2013; Sallam et al., 2020; Su, 2021; Sudhir & Unnithan, 2018; Sudhir & Unnithan, 2019; Zhu & Zhao, 2016) suggest that individuals' emotional states play an important role in how they respond to misinformation. For instance, Sudhir and Unnithan (2019) suggest that both positive and negative emotional states can trigger the sharing behavior of market-related misinformation. Han et al. (2020) found that people expressing high levels of anger were more likely to spread misinformation about COVID-19. Furthermore, research indicates that emotional state can interact with individuals' cognitive processes and motivations to shape people's beliefs and response to misinformation on social media. For instance, an anger state promotes the process of motivated reasoning, which further fuels belief in political misinformation (Weeks, 2015). An anxious state is positively correlated with people's emotional regulation motivation, which motivates people's engagement in sharing marketplace misinformation (Sudhir & Unnithan, 2018).

3.1.4. Context-related factors

The features of the entire information environment and social context, including normative cues, channel attributes, social network features, and debunking activities, wherein the dissemination of misinformation operates can affect the process of the spread of misinformation.

Normative cues. According to the theory of planned behavior, social norms play a pivotal role in shaping users' behaviors (Ajzen, 1991). In the social media context, various normative cues, such as the number of forwards and likes a post received, have been identified as factors influencing social media users' attitudes and behaviors toward misinformation posts (Bodaghi & Oliveira, 2020; Buchanan, 2020; Lee & Oh, 2017; Liu et al., 2013; Safarnejad et al., 2020). Additionally, comments from other users for a misinformation post showed a significant impact on individuals' intentions to share misinformation (Colliander, 2019; Wijenayake et al., 2020). On social media platforms, the source of misinformation is usually unknown or unfamiliar; thus, the liking of misinformation posts by reputable or familiar users would increase the misinformation's perceived credibility (Mena, Barbe, & Chan-Olmsted, 2020).

Channel attributes. The characteristics of social media platforms, such as the low entry barrier, thin-slice format, and interaction affordance, play a role in changing users' behavior, such as responding to misinformation posts (Allcott & Gentzkow, 2017; Di Domenico, Sit, et al., 2021). Some technological attributes of social media platforms, such as media richness, synchronicity, and recommendation mechanism, were also found to be correlated with the spread of misinformation (Lee & Shin, 2021; Oh, Gupta, Agrawal, & Rao, 2018; Peng & Yang, 2019). Compared with other communication channels (e.g., traditional media, online news websites, and face-to-face communication), social media was found to be the major channel contributing to the spread of food-related misinformation in China (Chang, Schulz, Jiao, Yu, & Yang, 2022; Wang et al., 2020). Additionally, different social media platforms show different susceptibility to misinformation. For example, an empirical study found that it was easier and faster to share misinformation on WhatsApp, followed by Facebook (Igwebuike & Chimuanya, 2021). Weibo was found to be the main social media platform that promotes the spread of misinformation in China (Guo & Zhang, 2020).

Social network features. The topology attributes of a social network reflect how people connect on the social media platform, including social tie strength, network density, social homogeneity, and social reinforcement, which play important roles in the spread of

misinformation. Specifically, the social tie strength is the level of intimacy of the social relationship or the number of overlapping friends between two individuals (Steffes & Burgee, 2009). The social tie strength has been shown to have a positive effect on misinformation belief (Oh et al., 2018) and the dissemination of misinformation (Apuke & Omar, 2020a; Huo, Lin, Liu, & Fang, 2019; Wang & Liu, 2019). Social network density, often considered as an important aspect of social capital, is found to advance information diffusion within the network. Recent research has revealed the positive effects of social network density (Burbach, Halbach, Ziefle, & Calero Valdez, 2019; Huang, Chen, & Ma, 2021; Qian, Tang, Zhang, & Zheng, 2015) on the spread of misinformation. Social homogeneity measures the similarity of interests or views among individuals. Many works have confirmed that social homogeneity is the primary driver of misinformation propagation (Choi, Chun, Oh, Han, & Kwon, 2020; Del Vicario et al., 2016; Kaligotla, Yücesan, & Chick, 2022; Mortimer, 2017). Furthermore, studies on misinformation have suggested that online users sharing similar interests or views tend to gather and eventually form a homogeneous cluster known as an echo chamber (Choi et al., 2020; Del Vicario et al., 2016; Mortimer, 2017). The core echo chamber with high degrees plays a crucial role in spreading misinformation (Choi et al., 2020). Social reinforcement is defined as the phenomenon wherein individuals' information behaviors can be affected by several pieces of information from their neighbors. As indicated by many existing studies (Xia et al, 2018; Wang, Deng, Huang, & Zhao, 2012; Wang, Deng, Xie, et al., 2012; Xu, Mei, Yang, & Kong, 2019; Ma, Li, & Tian, 2016; Huo & Chen, 2020; Huo, Chen, Xie, Liu, & He, 2021), social reinforcement is one of the key factors in the spread of misinformation on social networks. This means individuals are more likely to spread the misinformation if they have received the misinformation from their neighbors many times.

Debunking activities. The spread of misinformation is significantly affected by debunking activities. For example, the diffusion of debunking messages (Burel et al., 2020; Chua & Banerjee, 2018; Chung & Kim, 2021), presence of fact-checked labels (Ardèvol-Abreu, Delponti, & Rodríguez-Wangüemert, 2020; Calero Valdez & Ziefle, 2018; Hameleers et al., 2020; Mena, 2020; Nekmat, 2020), and users' active engagement in correction action (Talwar et al., 2020). Additionally, Jiang, Gao, and Zhuang (2021) explored the interplay mechanism between rumor-spreading and debunking processes and found that the debunking activity could have a significant influence on the final scope of misinformation propagation. However, several studies (Lewandowsky, Ecker, Seifert, Schwarz, & Cook, 2012; Pennycook, McPhetres, et al., 2020) found that although exposing users to misinformation-debunking messages can be useful in the short term, it is risky in the long term, because people might forget the details of the corrective content to which they were briefly exposed and instead rely on the increased familiarity of a misinformation statement when forming opinions; this is known as the familiarity backfire effect of misinformation-debunking activities (Scheufele & Krause, 2019). Furthermore, the worldview backfiring effect (Trevors & Duffy, 2020) of misinformation-debunking activities might also inadvertently strengthen belief in the misinformation that is being corrected.

3.2. *Major solutions to combat the spread of misinformation on social media*

Various approaches were proposed to fight against the spread of misinformation on social media focusing on the four fundamental communication elements according to the SMCR communication model: source-based approaches (source-targeting), message-based approaches (message-targeting), network-based approaches (context-targeting), policy-based approaches (context-targeting), and education-based approaches (receiver-targeting). Fig. 6 illustrates the specific strategies and the number of related articles.

3.2.1. *Message-based approaches*

Message-based approaches, including fact-checking the veracity of content, labeling misinformation posts with warnings, and issuing corrective messages to refute the misinformation on social media, play an important role in combating the spread of misinformation after its emergence.

Fact-checking. Fact-checking is described as a process consisting of three major steps: (1) selecting claims to check, (2) collecting evidence, and (3) deciding the appropriate fact-check tag for the claim (Craig & Yousuf, 2013; Hassan, Yousuf, Mahfuzul Haque, A. Suarez Rivas J., & Khadimul Islam, 2019). A natural approach is to hire professional fact-checkers to conduct fact-checking, but professional fact-checking is a laborious process that cannot be applied to the tremendous number of posts generated on social media every day. Consequently, automatic fact-checking emerges, which aims to verify claims automatically by synthesizing stances from various sources regarding the claim and providing a fact-check tag to show the extent of its veracity (Figueira,

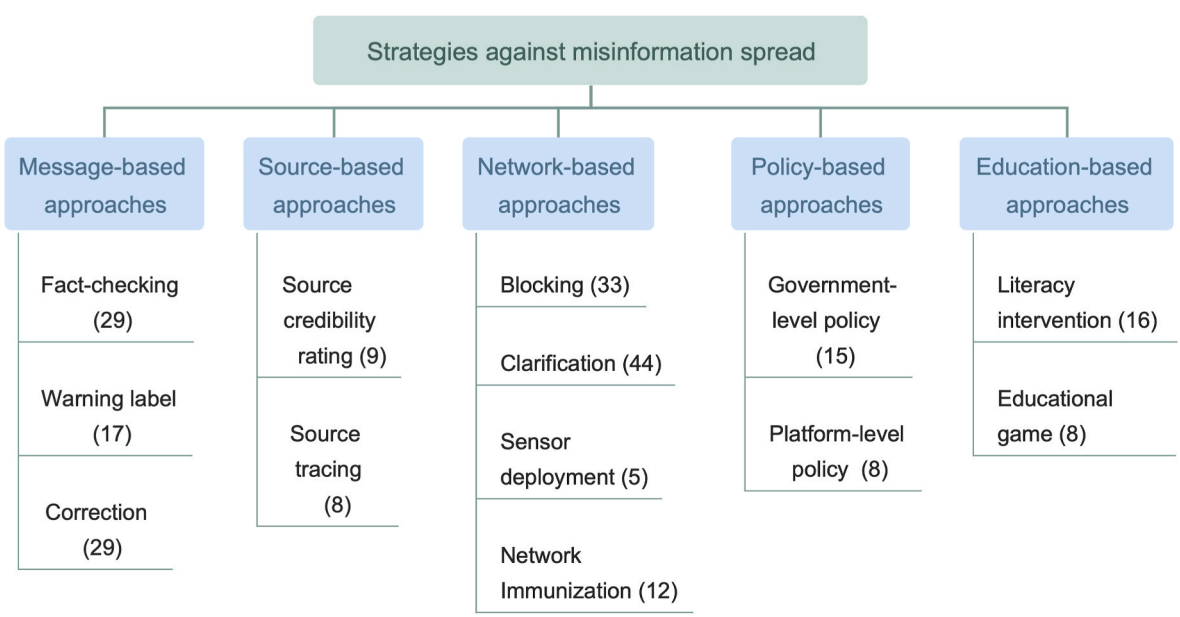

Fig. 6. Strategies against the spread of misinformation (201 related articles in total, 22 of which are about the combination of different types of strategies).

Guimaraes, & Torgo, 2019). Various technological solutions (Balmau et al., 2019; Guo, Schlichtkrull, & Vlachos, 2022; Jaroucheh, Alissa, Buchanan, & Liu, 2020; Karduni et al., 2018; Li, Dong, & Yang et al., 2020; Marcondes, Almeida, Durães, & Novais, 2020; Pinheiro, Cappelli, & Maciel, 2016; Pourghomi, Safieddine, Masri, & Dordevic, 2017; Thomson et al., 2022) are proposed to automate fact-checking by applying natural language processing, machine learning, knowledge graph query, and block-chain techniques. Various fact-checking tools have been developed, such as Truthy (Ratkiewicz et al., 2011), Right-click Authenticate (Safieddine et al., 2016), and Check-it (Paschalides et al., 2019). Although automatic fact-checking tools can be faster than professional fact-checking, most of them are domain-specific and susceptible to failure in the rapidly changing information environment (Allen, Arechar, Pennycook, & Rand, 2021). A complement to professional and automatic fact-checking is crowd-sourced fact-checking (Coscia & Rossi, 2020; Flores-Saviaga & Savage, 2021; Li & Sakamoto, 2015; Mohd Ramlan, Abdullah, Karkonasasi, & Mousavi, 2019). For instance, Mohd Ramlan et al. (2019) proposed a prototype based on a Twitter-like environment by adding crowd-sourced rating features.

Warning label. A general warning label attached to a social media post, which warns people that the post is incorrect, can decrease people's belief in misinformation and their decision to share it (Andı & Akesson, 2020; Ardèvol-Abreu et al., 2020; Clayton et al., 2020; Figl, Kießling, Rank, & Vakulenko, 2019; Mena, 2020; Pennycook, Bear, et al., 2020). Warnings attached with explanations of why the post had been marked or links providing additional information are found to be more effective (Kirchner & Reuter, 2020; Moravec, Kim, & Dennis, 2020). Furthermore, the study by Moravec et al. (2020) indicates that the combination of a stop icon signifying the post is fake and a link explaining why the post is fake is most effective. Nevertheless, although warnings may be able to fight misinformation, they are also criticized for unintentional adverse effects. For example, Clayton et al. (2020) found that general warning labels of misinformation shown to people before news exposure reduce individuals' trust in either real or false news on social media. The study by Ahsan, Kumari, and Sharma (2019) also highlights a drawback of misinformation warnings, that is, unlabeled misinformation posts would be perceived as more accurate, which is known as the "implied truth effect". Furthermore, invalid misinformation warnings would taint the truth and lead individuals to discard accurate information (Freeze et al., 2021).

Corrective communication. As mentioned earlier, correcting misinformation can backfire because at times, it may render the misinformation content more familiar to social media users. Therefore, it is critical to understand the different practices for correcting misinformation and their effectiveness. The most common type of corrective communication strategy is fact-based correction, which directly addresses the wrong claim in the misinformation and provides the alternative truth (Jin, van der Meer, Lee, & Lu, 2020; Van der Meer & Jin, 2020). Differently, logic-based correction focuses on undermining the logical flaws common in some types of misinformation (Vraga et al., 2019). One advantage of logic-based correction is that it might offer general protection against different types of misinformation that apply similar reasoning fallacies and strategies (Cook, Lewandowsky, & Ecker, 2017; Schmid & Betsch, 2019). The argument structure of correction also matters. For instance, a two-sided correction, which displays two opposing sides of claims while supporting one side by providing sufficient arguments and evidence, was found to be a promising strategy to address the negative impact of misinformation (Featherstone & Zhang, 2020; Ozturk, Li, & Sakamoto, 2015). Furthermore, corrective message is recommended to be delivered in a narrative style to explain a causal reasoning for the false claim of misinformation and offer new causal chains for individuals to replace their preexisting mental models (Lee, 2022; Shelby & Ernst, 2013). Additionally, several studies confirmed the effectiveness of corrective information delivered in humorous genres (Kim, Vraga, & Cook, 2021; Vraga et al., 2019). This effectiveness highlights how humorous communication can affect individuals' information processes, such as their degree of attention, and ultimately influence their attitudes toward the issue (Kim et al., 2021). Despite the correction content, the source of correction also plays an important role in debunking misinformation. For instance, broadcasting corrective messages through authoritative sources would enhance the effectiveness of correction practices (Chua, Cheah, Goh, & Lim, 2016; Margolin, Hannak, & Weber, 2018; Pal, Chua, & Goh, 2019; Vraga & Bode, 2017; Vraga et al., 2022). However, authoritative sources may not necessarily connect well with every social media user. Hence, self-organized or crowd-sourced correction practices demonstrate great potential (Arif et al., 2017; Li & Sakamoto, 2015; Malhotra, 2020; Zeng et al., 2019). For example, the study by Malhotra (2020) shows that misinformation-correcting posts from ordinary users appear to be more effective than those from official media accounts. Furthermore, with a growing trend of cross-platform information-seeking and -sharing behavior (Flores-Saviaga & Savage, 2021; Hunt, Wang, & Zhuang, 2020), cross-platform correction, which exposes people to consensus corrective messages across multiple social media platforms, is a promising development in combating misinformation (Hunt et al., 2020; Zhao, 2019).

3.2.2. *Source-based approaches*

Source-based approaches are considered alternatives or complementary to other message-based approaches (Pennycook & Rand, 2019b). For example, source credibility ratings based on evaluating past articles published by the sources may affect the extent to which users tend to share social media posts (Kim & Dennis, 2019), thereby avoiding the delay of message-based approaches (Kim, Moravec, & Dennis, 2019). Additionally, by using crowdsourcing to evaluate the credibility of news sources (rather than individual stories) and then using ratings to adjust social media ranking algorithms, users are more likely to engage with content originating from news sources that are broadly trusted by the crowds (Epstein, Pennycook, & Rand, 2020; Pennycook & Rand, 2019b; Faraon, Jaff, Nepomuceno, & Villavicencio, 2020; Hughes et al, 2014). Some research also explored technological approaches to evaluate the credibility of news sources (Chen & Freire, 2020; Huckle & White, 2017; Saad et al., 2019; Schwartz & Overdorf, 2020) or keep track of the source of misinformation from a set of social media posts (Nguyen, Nguyen, & Thai, 2012; Maryam & Ali, 2019; SuthanthiraDevi & Karthika, 2019; Khobragade, Landge, Kalani, & Patil, 2019; Qu, Zhao, Fu, Wang, & Xu, 2020; Louni & Subbalakshmi, 2018; Shuhud et al., 2017; Fraga-Lamas & Fernández-Caramés, 2020). For example, Huckle and White (2017) proposed a blockchain-based solution that validated the source of any digital media, such as images used out of context to mislead. To leverage the underlying correlation between the task of classifying a claim (e.g., news or messages) as truth or misinformation and the task of tracing the source of the claim, the use of a computational framework named SourceCR was proposed to implement the joint inference on truth/falsehood and their sources (Qu et al., 2020).

3.2.3. *Network-based approaches*

Network-based approaches aim to minimize the number of users accepting (or infected by) misinformation in the social network and are usually conducted through computational techniques and numerical simulations (Zareie & Sakellariou, 2021). The methods can be broadly divided into four categories: blocking strategy, clarification strategy, sensor deployment, and network immunization.

Blocking strategy. The principle of blocking strategy is to determine a set of nodes and/or edges (i.e., social media accounts and relationships between accounts) to be blocked or removed from the social network (Alorini, Rawat, & Alorini, 2020; Hosni, Li, & Ahmad, 2019; Hosni & Li, 2020; Ibrahim, Hefny, & Hassanien, 2016; Litou, Kalogeraki, Katakis, & Gunopulos, 2017; Ma et al., 2016; Maryam & Ali, 2020; Mehta, Mukhoty, & Gupta, 2016; Pham, Phu, & Hoang, 2018; Pham, Duong, Pham, Bao, & Nguyen, 2019; Pham et al., 2019; Pham, Nguyen, Nguyen,

Pham, & Nguyen, 2020; Wen et al., 2014; Xian et al., 2020; Yan et al., 2018; Yan et al., 2019; Yuan & Sun, 2020; Zhang, Wang, & Qin, 2019; Zhang, Bao, Niu, et al., 2019; Zheng & Pan, 2018; Zhu, Ni, & Wang, 2020). In blocking strategies, identifying the most influential spreaders in the social network is crucial. For example, in the study by Dey and Roy (2017), the identification of influential nodes is based on three centrality measures, including the degree distribution, betweenness and closeness centrality; the results indicate that the size of ultimately infected nodes is the smallest when nodes with high closeness centrality are blocked. Moreover, in some studies (Pham et al., 2018, 2019; Zheng & Pan, 2018), the goal is to minimize the budget of blocking, that is, to find the smallest set of nodes and/or edges whose removal from the network can bring the decrease of misinformation propagation larger than a threshold. However, in practice, blocking strategies may affect social media users' experiences, which may result in users responding negatively and falling away (Wang et al., 2017), or it may be regarded as a violation of freedom of expression (Hosni et al., 2019). Therefore, some research also considers the influence of blocking time and user experience (Hosni et al., 2019; Tripathi & Rao, 2022; Wang et al., 2017).

Clarification strategy. The clarification strategy focuses on selecting a set of seed nodes to conduct a truth campaign, that is, to distribute a clarification message to debunk the misinformation and raise users' awareness. In the selection of nodes to initiate truth campaigns, properties of network topology are widely applied (Choi, Moon, Shin, & Yi, 2016; Fan et al., 2013; Ghoshal, Das, & Das, 2019; Ghoshal, Das, & Das, 2021; Hosni, Li, & Ahmed, 2018; Hosni, Li, & Ahmad, 2020; Li, Li, Meng, Song, & Jiang, 2020; Litou, Kalogeraki, Katakis, & Gunopulos, 2016; Nguyen, Yan, & Thai, 2013; Ni, Guo, Huang, & Wu, 2020; Zhang, Zhang, Li, & Thai, 2015; Zhu & Zhao, 2017). For example, leveraging both the disjoint and overlapped community structures of the social network makes the process of selecting seed nodes simpler, faster, and independent of the distribution of misinformed nodes (Ghoshal et al., 2021). Moreover, user characteristics and behavior are considered when selecting the seed nodes (Fan et al., 2014; Mo, Tian, Wang, & Peng, 2019; Song et al., 2017; Tong et al., 2018; Wu et al., 2018), with the expectation that user preferences and behavior may allow for more elaborate differentiation of nodes than purely structure-based methods. For example, considering the mobility of users, the classical SIR model is extended to simulate the misinformation diffusion process, and a set of mobile nodes that can move to neighbor nodes are chosen to propagate the correct information (Mo et al., 2019). However, compared with network structure information, user-related information is more difficult to access in the real world, which makes the structure-based methods more widely studied and applied.

Sensor deployment. The sensor/monitor deployment strategy aims to place monitors near (identified or suspected) misinformation sources and filter the information spread by these sources. Computer-aided social media accounts or real paid user accounts are employed to operate monitoring; their role is to filter the information they receive and block what they consider to be misinformation. The number of monitors should be as few as possible because they may be expensive to deploy and maintain, and the distance of monitors should be as close as possible to the suspected sources so as to limit their reachability to other nodes (Zhang, Alim, Thai, & Nguyen, 2015). The network monitors can be determined by finding the partition of the graph that has the largest transmission probability across the partitions and a maximum of k nodes on the side of (identified or suspected) misinformation sources (Amoruso et al., 2020). To better approximate a realistic setting, the distributions of transmission time delays have also been considered (Spinelli, Celis, & Thiran, 2017). Another possible sensor placement method is based on detecting communities in a network and finding the articulation points within those communities, and then monitors will be placed at these articulation points (Maryam & Ali, 2020). Unlike the previous two types of network-based strategies, the sensor placement strategy may help restrict the spread distance of misinformation even before it is detected. However, when the network topology changes, the sensor placement must be updated accordingly. Additionally, in real online social networks, asking a user to act as a monitor might be rather difficult, which means monitors are not suitable to be placed everywhere.

Network immunization. In the field of complex networks and rumor dynamics, researchers apply epidemic propagation models (e.g., Moreno, Nekovee, & Pacheco, 2004; Pastor-Satorras, Castellano, Van Mieghem, & Vespignani, 2015; Zhao et al., 2013) to study the spread of misinformation and adapt epidemic immunization strategies (e.g., Madar, Kalisky, Cohen, Ben-avraham, & Havlin, 2004; Pastor-Satorras & Vespignani, 2002; Wang et al., 2009) to fight against the spread. Such an immunization strategy usually contains three steps. First, select a set of nodes to be vaccinated. Second, determine max inter vaccination time. Third, apply vaccination to selected individuals at the time interval determined in Step 2. Misinformation researchers in this direction explore how to identify the network nodes that will be given the vaccination. The most frequently studied immunization strategies are random immunization and targeted immunization. Random immunization strategy randomly selects a set of nodes to give them knowledge about how to resist the spread of misinformation, whereas targeted immunization strategy selectively identifies a few key nodes to educate them about the negative impact of the spread of misinformation. Some research compared the efficiency of random immunization, targeted immunization, and their combination in online social networks with different topology characteristics (Hosseini, Abdollahi Azgomi, & Torkaman Rahmani, 2015; Singh & Singh, 2012a, 2012b). However, random and targeted immunizations have many difficulties in practice. For example, random immunization requires immunizing a large proportion of a network, whereas targeted immunization requires complete information about the network (Bao, Niu, Yi, & Xue, 2014). To overcome these disadvantages, several improved strategies were proposed based on random and targeted immunizations (Singh et al., 2013; Bao et al., 2014; Zhao et al., 2015; Fan, Wu, Xing, & Lee, 2016; Santhoshkumar and Dhinesh Babu, 2019). For example, Bao et al. (2014) propose an immunization strategy concerning maximum spanning tree, which achieves higher immunity performance and better maintaining of network connectivity than random and targeted immunizations. Santhoshkumar and Dhinesh Babu (2019) propose an approach to repeatedly immunize nodes with high spreading ability in specific interval time.

To overcome the shortcomings of the aforementioned network-based strategies and exploit their advantages, researchers further explored the compound methods of different strategies working together (Amoruso et al., 2020; Hosni et al., 2019; Lian, Liu, & Dong, 2020; Maryam & Ali, 2020; Wang et al., 2021; Wen et al., 2014; Yuan & Sun, 2020; Zhao et al., 2019). For instance, Hosni et al. (2019) propose a dynamic approach for rumor influence minimization, which finds a compromise between node blocking and truth campaign. Additionally, the isolation-conversion (IC) method is proposed, which aims to isolate influential misinformation spreaders and then convert them into truth spreaders (Wen et al., 2014; Zhao et al., 2019). The IC method saves time and social capital because it utilizes the spontaneous misinformation-debunking behavior of misinformation spreaders (Wang et al., 2021). However, these models usually assume that individuals' susceptibility to misinformation obeys specific mathematical distributions, which makes it difficult to verify the appropriateness of these models in real cases.

3.2.4. Policy-based approaches

In recent years, we have witnessed an increasing attention focus on policy- and law-based solutions to prevent the threat of misinformation. The increased policy interest in misinformation benefits the opening up of important and fundamental issues that need to be handled to secure a healthy and sustainable information ecosystem on the internet (Bechmann, 2020). Generally, these policies can be divided into two levels: government level (Alshible, 2020; Alvarez-Risco et al., 2020; Bali & Desai, 2019; Bechmann, 2020; Daud, 2020; Daud & Zulhuda, 2020;

Helberger, 2020; Kabha, Kamel, Elbahi, & Narula, 2019; Marsden, Meyer, & Brown, 2020; Martynov & Bundin, 2020; Neo, 2020; Radu, 2020; Rodrigues & Xu, 2020; Tan & Teng, 2020) and platform level (Anderson, 2018; Daud, 2020; Iosifidis & Nicoli, 2020; Rochlin, 2017; Rodrigues & Xu, 2020; Zeng et al., 2017).

Government-level policy. The apparent threat of misinformation has urged several national governments to enact laws and policies to combat misinformation. For example, Peru is the first Latin American country to threaten imprisonment for those who create and/or propagate misinformation to benefit themselves or injure others (Alvarez-Risco et al., 2020). With the enactment of the "Anti-Fake News Act", Malaysia sanctions any act of "creating, offering, publishing, printing, distributing, circulating, or disseminating any fake news or publication containing fake news" (Daud, 2020). To curtail the circulation of misinformation during COVID-19, the Chinese and Indian governments implemented several new anti-misinformation policies (Rodrigues & Xu, 2020). Moreover, several studies have compared the effectiveness of these national laws and policies. For example, the laws and policies in the United Arab Emirates (UAE) and China on managing misinformation on social media are found to be more effective than those in Jordan and the United Kingdom because of the presence of clearer, comprehensive, and explicit legal frameworks in the UAE and China (Alshible, 2020; Kabha et al., 2019).

Platform-level policy. To act against the serious consequences of misinformation spread, dominant social media platforms have also implemented policies and rules of use to regularize users' information behaviors. For example, Facebook's Community Guidelines state that any user who publishes or shares inaccurate, misleading, or fake information to collect likes, followers, or shares will be suspended (Daud, 2020; Iosifidis & Nicoli, 2020; Rochlin, 2017). YouTube bans any deceptive practices, such as fake news, spam, and scams (Daud, 2020). Since 2020, WeChat, Tik Tok, and Weibo (mainstream social media platforms in China) have implemented various policy campaigns against COVID-19-related misinformation (Rodrigues & Xu, 2020). Twitter has not proposed any clear policies to cope with misinformation but it is clearly against impersonation, spam, and bots (Anderson, 2018; Daud, 2020).

The job of regulating misinformation should not rely solely on national governments, neither should the social media platforms be fully responsible for regulating themselves. Instead, several studies (Marsden et al., 2020; Martynov & Bundin, 2020) have suggested the need to strengthen and intensify effective co-regulation between governments and social media platforms to deter the spread of misinformation. For example, social media platforms develop mechanisms individually or collectively to regulate their own users, which, in turn, must be supported by local governments and legislatures, which also guarantee and monitor their effectiveness (Marsden et al., 2020).

3.2.5. *Education-based approaches*

According to a report issued by the Open Society Institute, education is regarded as the best all-round solution for combating the phenomenon of misinformation spread because it has fewer drawbacks and more possibilities of being adopted in different situations (Gilchrist, 2018; Lessenski, 2018). Education-based approaches focus on how to promote healthier behavior among social media users to minimize the act of spreading misinformation, such as online guidance, mini-courses, and lectures, which aim to improve users' general information literacy (Bangani, 2021; Dib, Mayaud, Chauvin, & Launay, 2022; Guo & Huang, 2021; Lim & Tan, 2020; Musgrove, Powers, Rebar, & Musgrove, 2018; Tsipursky et al., 2018; Walker, 2019; Young et al., 2021) or specific cognitive skills (Admoko, Suprapto, Deta, Achmadi, & Hariyono, 2021; Athar, Pratama, & Mujtahid, 2019; Musgrove et al., 2018), and games or gamification elements designed to raise users' awareness or ability to discern online misinformation (Almaliki, 2019; Basol, Roozenbeek, & Van der Linden, 2020; García-Ortega and García-Avilés, 2021; Maertens, Roozenbeek, Basol, & Van der Linden, 2021; Roozenbeek & Van Der Linden, 2019a; Roozenbeek and Van der Linden, 2019b; Urban et al., 2018; Yang et al., 2021).

Information literacy interventions are designed to improve users' skills and knowledge to locate, understand, evaluate, and use information effectively (ACRL, 2000), and the application of such interventions might help users build the skills needed to combat misinformation. For example, faculties and librarians utilize the Association of College and Research Libraries Framework (ACRL, 2016) and various LibGuide applications (tools embedded in course and library websites to create and organize guidelines) to teach information literacy and critical thinking skills (Musgrove et al., 2018). Various teaching methods are used to promote information literacy education, such as live-stream lectures, videos of mini-courses, and massive open online courses deployed during the pandemic (Yang et al., 2021).

Although most work has used relatively passive educational interventions, game-based educational interventions are found to be promising in terms of engaging people in being more actively involved in combating the spread of misinformation. For example, Roozenbeek and Van Der Linden (2019a; 2019b) devised the "Bad News Game" grounded in psychological inoculation theory, wherein preemptively exposing people to weakened doses of the tactics commonly applied in the production of online misinformation helps enhance cognitive immunity when encountering real misinformation. Specifically, in the "Bad News Game", players adopt the roles of misinformation creators whose goal is to select appropriate posts applying various misinformation techniques to gain as many followers as possible while maintaining a sufficiently high level of credibility. Experimental studies (Basol et al., 2020; Maertens et al., 2021; Roozenbeek & Van der Linden, 2019b) demonstrated that the process of active inoculation triggered by the "Bad News Game" significantly reduced participants' perception of the reliability of misinformation. Additionally, gamification elements (e.g., points, levels, and badges) are advocated to be incorporated into the design process of social media platforms to build a more misinformation-aware environment (Almaliki, 2019). The study by Yang et al. (2021) confirmed the effectiveness of a digital game-based intervention in improving young adults' information discernment skills. Although educational games designed to counter the spread of misinformation are becoming prevalent, there is a dearth of research that systematically examines their effectiveness (Yang et al., 2021).

4. Discussion and future directions

To our knowledge, this work is the first effort to offer a systematic overview of both the factors affecting the spread of misinformation and strategies against misinformation in the social media ecosystem. The results of our analysis of these research articles point to several research gaps in both aspects.

4.1. *Factor-related research gaps*

In the reviewed articles, we found that source-, message-, and context-related factors that contribute to the viral spread of misinformation on social media have been investigated extensively. However, many important receiver-related factors are still largely unexplored; one of them being the receiver's persuasion knowledge. According to the persuasion knowledge model developed by Friestad and Wright (1994), individuals' topic knowledge, agent knowledge, and persuasion knowledge affect how they respond to persuasion attempts. In the context of persuasion attempts through misinformation, topic knowledge is the understanding of the issue or event mentioned in the misinformation content; agent knowledge is the belief regarding the traits of the misinformation source or sender; and persuasion knowledge includes the comprehension of the persuasion tactics that may be used to persuade people to believe in the misinformation and spread it, effectiveness of those tactics in the social media environment, underlying goals behind the origin and spread of misinformation, and coping

strategies when exposed to misinformation (Chen & Cheng, 2019; Xiao, 2021). Currently, most existing research focuses on investigating the influence of social media users' topic knowledge and agent knowledge on misinformation-spreading behaviors. It remains largely unknown how individuals' persuasion knowledge plays a role in the process. Although previous works (Chen, Xiao, & Mao, 2021; Zhou et al., 2021) have identified the persuasive features of misinformation content on social media, will the high level of individuals' persuasion knowledge raise their awareness of the persuasion attempt behind misinformation and decrease their tendency to spread misinformation on social media? If so, how to effectively increase social media users' persuasion knowledge? These are example questions that should be explored in the future.

We also notice that studies establishing the interaction effects between receiver-related and other factors are still scarce. As presented in our review of the literature, the crucial role of receiver-related factors in the process of misinformation spread has been well established. It helps introduce important insights regarding the moderating and/or mediating effect of receiver-related factors when researchers investigate the effects of other factors. For example, by establishing the significant mediating effect of an individual's worry about COVID-19 on the relationship between peer communication (contextual factor) and misinformation sharing, the study by Luo, Wang, Guo, and Luo (2021) indicates that discussing COVID-19 with friends too frequently would increase an individual's worry and further induce misinformation sharing on social media. Additionally, examining the interaction effects among receiver-related factors may disentangle the inconsistency of previous findings. For instance, regarding whether the personality of neuroticism predicts individuals' vulnerability to misinformation, Chen (2016) and Buchanan and Benson (2019) generate contradictory results. A closer look at these two studies shows that they were conducted among different population groups: the research by Chen (2016) used student samples from Singapore, whereas the research by Buchanan and Benson (2019) was conducted among participants from 44 different countries and whose ages ranged from 16 to 80 years. It is possible that the effect of receivers' personality traits on their tendency to share misinformation on social media may have been mediated by their age and culture, thereby causing the discrepancy in the results of the two studies.

4.2. Strategy-related research gaps

As detailed in the previous section, we identified five broad categories of solutions that have been explored to combat misinformation on social media, namely, message-, source-, network-, policy-, and education-based approaches. Furthermore, we noticed that several areas are largely overlooked in the research effort of developing such solutions: the explainability of message-, source-, and network-based technical solutions, overlooking of individuals' motivation and emotional state in the design of educational interventions, and need to verify the effectiveness of proposed strategies in real cases. We explain each of these aspects as follows.

To deter the rapid spread of misinformation, an array of high-performance fact-checking systems based on advanced Artificial Intelligence (AI) techniques were devised that leverage the message-, source-, and network-related characteristics of misinformation spread. However, it is no longer considered to be sufficient to have a system/model that just shows high performance in identifying potential misinformation (Mishima & Yamana, 2022; Szczepański et al., 2021). The ability to explain why a piece of message is detected as misinformation is particularly important in practical situations. Consider this scenario: certain crime-related information is classified as misinformation by the AI-based fact-checking system, the police or journalist cannot simply inform the public to discard it. Instead, they are expected to explicitly interpret why it is fake. Indeed, the lack of understanding regarding the decision-making processes behind the AI-based systems would impair users' trust in the result provided by the systems (Adadi & Berrada, 2018; Mohseni et al., 2020; Ribeiro, Singh, & Guestrin, 2016). Recently, explainable artificial intelligence (XAI) has gained increasing attention in the field of fact-checking and misinformation detection (Yang et al., 2019; Shu et al., 2019; Lu & Li, 2020; Mohseni et al., 2021; Ayoub, Yang, & Zhou, 2021; Szczepański et al., 2021; Chien, Yang, & Yu, 2022). However, these studies still focus on evaluating the performance of the system in predicting misinformation, except for a few that evaluate the explainability of the system and the impact of generated explanations on social media users (Chien et al., 2022; Mohseni et al., 2021; Yang et al., 2019). We envision that XAI-based fact-checking and misinformation detection is a future trend in combating the spread of misinformation.

Education-based interventions, such as information literacy education and educational games, have been highly recommended in recent years. Most of these solutions aim at improving individuals' knowledge levels and cognitive skills in identifying misinformation on social media. However, in some situations, the truthfulness of information may be less important than goals such as socializing, self-promoting, and protecting a prior belief. Thus, individuals may not only lack the knowledge and cognitive skills to discern misinformation but also the motivation to engage in careful cognitive processes to evaluate the information (Scheufele & Krause, 2019). Interventions that motivate individuals to pursue the truth are therefore also important. A few related studies explore this design direction. For instance, Lapsley and Chaloner (2020) suggest that nurturing science identity among the general public is critical for enhancing people's motivation to pursue truth. Chinn, Barzilai, and Duncan (2020) argue for interventions that relate to students' experiences and encounters with science and media in daily life to promote their reflection on their epistemic aims and processes. Further empirical studies are needed to put these motivation-target interventions into practice and verify their effectiveness in different situations. Another oversight in the education-based approach is the influence of the individual's emotional state. Only a few education-based interventions or gamification tools that teach people to become more aware of their emotional states when encountering information on social media, even though it has been widely acknowledged in the literature that individuals' emotional states influence how they process and share misinformation online. One such intervention is proposed by Veletsianos et al. (2022), which is based on narrative techniques to teach people to reflect on their emotions when they browse social media content. Further research is particularly desired to put the emotional intervention proposed by Veletsianos et al. (2022) into practice in a real social media context. Additionally, we call for more educational interventions to be designed and examined that help people recognize and attend to their emotional states when processing and consuming online information.

Lastly, our review of these solutions-focused research articles implies that although an array of misinformation-countering strategies has been proposed, more empirical studies are needed to establish the effectiveness of these strategies. First, more social media field experiments are needed to examine whether and how misinformation-countering interventions affect social media users' actual information sharing decisions (e.g., Mosleh, Martel, Eckles, & Rand, 2021; Pennycook et al., 2021). How the related knowledge and information literacy skills that individuals have gained in curriculum or educational games transfer to the external contexts, i.e., if players learn to identify misinformation via gameplay, they should be able to do so in various related contexts in their daily interactions with social media and online news. How does the effectiveness of these interventions extend over time? How do they affect people's beliefs and behaviors in the long term? The evaluation of these interventions needs to consider their external validity and ecological validity and the long-term effects on individuals' lives. Second, because receivers' characteristics are found to be an important type of factors in misinformation spread, misinformation-countering strategies may need to be tailored or personalized accordingly. For instance, the effectiveness of different types of corrective communication are found to be contingent upon individuals' motivation for using social

media (Lee, 2022) and preexisting beliefs regarding the topic targeted with misinformation (Wang, 2021). One direction for exploration is to examine how the diverse receiver-related factors mentioned in Section 3.1.3, such as age, cognitive ability, and mood, moderate the effectiveness of different types of misinformation correction strategies. Third, the field of network-based strategies against misinformation has been widely explored, but the proposed models usually assume that the individuals' attributes and network structure obey specific mathematical distributions. This assumption makes it difficult to verify the appropriateness of these strategies in real cases. We expect future research to consider more dynamic elements of user state and network structure so as to capture the actual interactions in real-world social networks.

5. Conclusion

Although the deluge and significant impact of misinformation in social media have been recognized, a comprehensive review of the triggering factors and countermeasures to misinformation from diverse perspectives is lacking. Following the process of systematic literature review, we sourced papers published between January 2010 and September 2021, from the WoS and Scopus databases, and synthesized 423 eligible articles. Compared with review research mentioned in the introduction (e.g., Baptista & Gradim, 2020; Ecker et al., 2022; Pennycook & Rand, 2021), more factors were identified in this review (e.g., psychological cues of the message, personality traits of the receiver, and topology features of the social network), and these factors were categorized according to the four vital elements of information communication (i.e., source, message, context, receiver). Furthermore, strategies confronting the viral spread of misinformation were summarized from five perspectives: message, source, social network, policy, and education, providing what they are and how effective they are.

There are several limitations to this systematic review. First, this review might not have covered all the factors affecting the spread of misinformation and the strategies against misinformation on social media, given that some relevant articles might have applied different keywords and given our strict inclusion and exclusion criteria. Second, the journals and conference proceedings sourced were limited to the particular databases included in this study. Furthermore, our focus on peer-reviewed publications means that potentially relevant works in other formats (e.g., technical reports, policy documents, and preprints) could have been overlooked. Third, the policies and rules of different social media platforms to prevent the spread of misinformation described in Section 3.2.4 may now be outdated because policies changes regularly. Finally, because we excluded articles not published in English, we may have omitted important factors or strategies that have been explored but reported in other languages.

Notwithstanding these limitations, this review is still beneficial in providing a synthesis of diverse factors underpinning the spread of misinformation on social media and a general picture of countermeasures from different disciplines and perspectives. Based on the studies reviewed, we highlighted several research gaps that future research should address. Specifically, regarding the investigation of what contributes to the spread of misinformation, we encourage studies that examine the role of individuals' persuasion knowledge and the complex interplay between receiver-related and other factors based on a combination of self-reporting, behavioral, neural, and/or biophysical data. For research that explores strategies to combat misinformation in several areas, we speculate on the potential of using XAI-based solutions, consideration of individuals' psychological aspects, such as motivation and emotion in the design of educational interventions, and necessity of evaluating these strategies in real-world situations.

Declaration of interest statement

We have no conflict of interest.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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